

Energetics of the single DOF oscillator

MG3416–Advanced Structural Acoustics - Lecture #1

part B

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Outline

SDOF
energetics

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SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition
Stationary
evolutionary
response

Summary

1 SDOF oscillator (reminder)

- Notations
- Energetic quantities

2 Stationary excitation

- Stationary forced response
- Equipartition
- Stationary evolutionary response

Outline

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition
Stationary
evolutionary
response

Summary

1 SDOF oscillator (reminder)

- Notations
- Energetic quantities

2 Stationary excitation

- Stationary forced response
- Equipartition
- Stationary evolutionary response

Single DOF oscillator

Notations

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations

Energetic
quantities

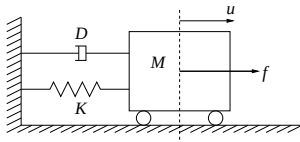
Stationary
excitation

Stationary
forced
response

Equipartition

Stationary
evolutionary
response

Summary



- The single degree-of-freedom (DOF) oscillator:

$$\begin{cases} M\ddot{u}(t) + D\dot{u}(t) + Ku(t) = f(t), & t \in \mathbb{R}, \\ u(0) = u_0, \\ \dot{u}(0) = v_0; \end{cases}$$

$M > 0$: mass, $D \geq 0$: damping, and $K > 0$: stiffness.

- Fundamental parameters:

- $\omega_p = \sqrt{\frac{K}{M}}$: the undamped natural (angular) frequency,
- $\xi_p = \frac{D}{2\sqrt{KM}}$: the critical damping rate,
- $\eta_p = 2\xi_p$: the loss factor.

Single DOF oscillator

Free response

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energetics

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SDOF
oscillator
(reminder)

Notations

Energetic
quantities

Stationary
excitation

Stationary
forced
response

Equipartition

Stationary
evolutionary
response

Summary

Definition

The free response $t \mapsto u^\ell(t)$ is the solution of:

$$\begin{cases} M\ddot{u}(t) + D\dot{u}(t) + Ku(t) = 0, & t \geq 0, \\ u(0) = u_0, \\ \dot{u}(0) = v_0. \end{cases}$$

- Let $\omega_D = \omega_p \sqrt{1 - \xi_p^2}$, then:

$$u^\ell(t) = e^{-\xi_p \omega_p t} \left(u_0 \cos \omega_D t + \frac{\xi_p \omega_p u_0 + v_0}{\omega_D} \sin \omega_D t \right).$$

- **Property:** if $D > 0$, $\lim_{t \rightarrow +\infty} u^\ell(t) = \lim_{t \rightarrow +\infty} \dot{u}^\ell(t) = 0$.

Single DOF oscillator

Forced response

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations

Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition

Stationary
evolutionary
response

Summary

Definition

The forced response $t \mapsto u^f(t)$ is the solution of:

$$M\ddot{u}(t) + D\dot{u}(t) + Ku(t) = f(t), \quad t \in \mathbb{R}.$$

■ Then:

$$u^f(t) = \int_{-\infty}^t \mathfrak{h}(t - \tau) f(\tau) \, \mathrm{d}\tau = \int_0^{+\infty} \mathfrak{h}(\tau) f(t - \tau) \, \mathrm{d}\tau,$$

where $\mathfrak{h} : \mathbb{R} \rightarrow \mathbb{R}$ is the impulse response function of the SDOF oscillator:

$$\mathfrak{h}(t) = \mathbb{1}_{[0, +\infty[}(t) \times \frac{1}{M\omega_D} e^{-\xi_p \omega_p t} \sin \omega_D t.$$

Single DOF oscillator

Frequency response function

SDOF
energetics

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SDOF
oscillator
(reminder)

Notations

Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition

Stationary
evolutionary
response

Summary

- $t \mapsto \mathbb{h}(t)$ is integrable and square integrable on $\mathbb{R}$, and its Fourier transform is:

$$\hat{\mathbb{h}}(\omega) = \int_{\mathbb{R}} e^{-i\omega t} \mathbb{h}(t) dt = \frac{1}{M(\omega_p^2 - \omega^2 + 2i\xi_p\omega_p\omega)}.$$

- $\omega \mapsto \hat{\mathbb{h}}(\omega)$ is integrable and square integrable on $\mathbb{R}$, and its inverse Fourier transform is:

$$\mathbb{h}(t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\omega t} \hat{\mathbb{h}}(\omega) d\omega.$$

- Usual quadratures:

$$\int_0^{+\infty} |\hat{\mathbb{h}}(\omega)|^2 d\omega = \frac{\pi}{2DK},$$
$$\int_0^{+\infty} \omega^2 |\hat{\mathbb{h}}(\omega)|^2 d\omega = \frac{\pi}{2DM}.$$

Single DOF oscillator

Impedance

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oscillator
(reminder)

Notations

Energetic
quantities

Stationary
excitation

Stationary
forced
response

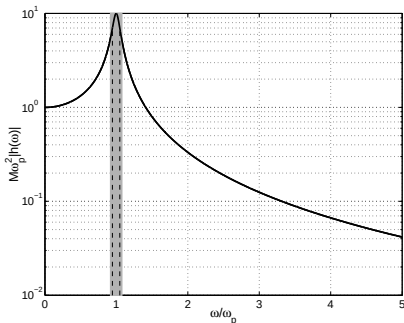
Equipartition

Stationary
evolutionary
response

Summary

- The **impedance** $\omega \mapsto Z(\omega)$ of the SDOF oscillator:

$$i\omega Z(\omega) = M(\omega_p^2 - \omega^2 + 2i\xi_p\omega_p\omega).$$



- $\omega'_p = \omega_p \sqrt{1 - 2\xi_p^2}$: the natural resonance frequency (for $0 < \xi_p < \frac{1}{\sqrt{2}}$);

- $b_e = \pi \xi_p \omega_p (1 - \xi_p^2)$: the **equivalent bandwidth**, s.t.

$$b_e |\hat{h}(\omega'_p)|^2 = \int_0^{+\infty} |\hat{h}(\omega)|^2 d\omega;$$

- $b_p \simeq \pi \xi_p \omega_p = \frac{\pi}{2} \Delta_p$, where $\Delta_p = \eta_p \omega_p$ is the **half-power bandwidth**.

Single DOF oscillator

Evolutionary response

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SDOF
oscillator
(reminder)

Notations

Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition

Stationary
evolutionary
response

Summary

Definition

The evolutionary response $t \mapsto u(t)$ is the solution of:

$$\begin{cases} M\ddot{u}(t) + D\dot{u}(t) + Ku(t) = f(t), & t \geq 0, \\ u(0) = u_0, \\ \dot{u}(0) = v_0. \end{cases}$$

■ Then:

$$\begin{aligned} u(t) = u_0 e^{-\xi_p \omega_p t} & \left(\cos \omega_D t + \frac{\xi_p}{\sqrt{1 - \xi_p^2}} \sin \omega_D t \right) \\ & + \frac{v_0}{\omega_D} e^{-\xi_p \omega_p t} \sin \omega_D t + \int_0^t \mathbb{h}(t - \tau) f(\tau) d\tau. \end{aligned}$$

■ **Property:** if $D > 0$, $\lim_{t \rightarrow +\infty} |u(t) - u^f(t)| = 0$.

Energetic quantities

Definitions

Definition

- The *kinetic energy*: $\mathcal{E}_c(t) = \frac{1}{2}M\dot{u}(t)^2$,
- The *potential energy*: $\mathcal{E}_p(t) = \frac{1}{2}Ku(t)^2$,
- The *mechanical energy*: $\mathcal{E}(t) = \mathcal{E}_c(t) + \mathcal{E}_p(t)$,
- The *dissipated power*: $\Pi_d(t) = D\dot{u}(t)^2$,
- The *input power*: $\Pi_{IN}(t) = f(t)\dot{u}(t)$.

- The instantaneous power balance reads:

$$\dot{\mathcal{E}}(t) = \Pi_{IN}(t) - \Pi_d(t).$$

- It is subsequently specialized to the free, forced and evolutionary responses of the single DOF oscillator.

Energetic quantities

Free response

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(reminder)

Notations

Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition

Stationary
evolutionary
response

Summary

- The free response mechanical energy when $\xi_p \ll 1$:

$$\mathcal{E}^\ell(t) \simeq \mathcal{E}_0 e^{-\eta_p \omega_p t},$$

where $\mathcal{E}_0 = \frac{1}{2} M v_0^2 + \frac{1}{2} K u_0^2$.

- The power balance integrated between 0 and $t > 0$:

$$\mathcal{E}_0 = \mathcal{E}_d^\ell(t) + \mathcal{E}^\ell(t),$$

hence $\mathcal{E}_d^\ell(\infty) = \mathcal{E}_0 = \int_0^{+\infty} \Pi_d^\ell(t) dt$.

Energetic quantities

Forced response

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)
Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition
Stationary
evolutionary
response

Summary

- **Data:** square integrable (finite energy) excitation with limited bandwidth,

$$|\hat{f}(\omega)| \leq C \quad \forall \omega \in \mathbb{R}; \quad \hat{f}(\omega) = 0 \quad \forall \omega \notin [-\omega_f, \omega_f].$$

- Then $\lim_{|t| \rightarrow +\infty} f(t) = 0$, and consequently **if $D > 0$:**

$$\lim_{|t| \rightarrow +\infty} u^f(t) = \lim_{|t| \rightarrow +\infty} \dot{u}^f(t) = 0.$$

- The power balance integrated between $-\infty$ and t :

$$\begin{aligned} \mathcal{E}^f(t) &= \int_{-\infty}^t \Pi_{\text{IN}}(\tau) \, d\tau - \int_{-\infty}^t \Pi_{\text{d}}^f(\tau) \, d\tau \\ &= \mathcal{E}_{\text{IN}}(t) - \mathcal{E}_{\text{d}}^f(t) \end{aligned}$$

since $\mathcal{E}^f(-\infty) = 0$. But $\mathcal{E}^f(+\infty) = 0$ as well, hence:

$$\mathcal{E}_{\text{IN}}(+\infty) = \mathcal{E}_{\text{d}}^f(+\infty).$$

Energetic quantities

Equipartition in the forced response

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SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response

Equipartition

Stationary
evolutionary
response

Summary

■ Hypotheses (wideband excitation):

■ i $\xi_p \ll 1;$

■ ii $\omega \mapsto \hat{f}(\omega)$ varies slowly on $[-\omega_f, \omega_f]$ and $\omega_f \gg \omega_p$.

■ Then:

$$\begin{aligned}\int_{\mathbb{R}} \mathcal{E}_c^f(t) dt &= \frac{M}{4\pi} \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 |\hat{f}(\omega)|^2 d\omega && \text{(Plancherel)} \\ &\simeq \frac{M}{4\pi} |\hat{f}(\omega_p)|^2 \int_{-\omega_f}^{\omega_f} \omega^2 |\hat{h}(\omega)|^2 d\omega && \text{(using (ii))} \\ &\simeq \frac{M}{2\pi} |\hat{f}(\omega_p)|^2 \int_0^{+\infty} \omega^2 |\hat{h}(\omega)|^2 d\omega && \text{(using (i))} \\ &= \frac{|\hat{f}(\omega_p)|^2}{4D}\end{aligned}$$

- Likewise $\int_{\mathbb{R}} \mathcal{E}_p^f(t) dt \simeq \frac{|\hat{f}(\omega_p)|^2}{4D}$, independently of the mass or the stiffness.

Energetic quantities

Equipartition in the forced response (extended proof)

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations

Energetic
quantities

Stationary
excitation

Stationary
forced
response

Equipartition

Stationary
evolutionary
response

Summary

$$\begin{aligned}\int_{\mathbb{R}} \mathcal{E}_c^f(t) dt &= \frac{M}{2} \int_{\mathbb{R}} (\dot{u}^f(t))^2 dt \\&= \frac{M}{4\pi} \int_{\mathbb{R}} |\hat{u}^f(\omega)|^2 d\omega \quad (\text{Plancherel } \int_{\mathbb{R}} |u(t)|^2 dt = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{u}(\omega)|^2 d\omega) \\&= \frac{M}{4\pi} \int_{\mathbb{R}} |i\omega \hat{u}^f(\omega)|^2 d\omega \quad (\text{Fourier transform } \hat{u}(\omega) = i\omega \hat{u}(\omega)) \\&= \frac{M}{4\pi} \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 |\hat{f}(\omega)|^2 d\omega \quad (u^f(t) = \int_{-\infty}^t h(t-\tau) f(\tau) d\tau) \\&\quad \implies \hat{u}^f(\omega) = \hat{h}(\omega) \hat{f}(\omega) \\&\simeq \frac{M}{4\pi} |\hat{f}(\omega_p)|^2 \int_{-\omega_f}^{\omega_f} \omega^2 |\hat{h}(\omega)|^2 d\omega \quad (\text{using (ii)}) \\&\simeq \frac{M}{2\pi} |\hat{f}(\omega_p)|^2 \int_0^{+\infty} \omega^2 |\hat{h}(\omega)|^2 d\omega \quad (\text{using (i)}) \\&= \frac{|\hat{f}(\omega_p)|^2}{4D} \quad (\int_0^{+\infty} \omega^2 |\hat{h}(\omega)|^2 d\omega = \frac{\pi}{2DM})\end{aligned}$$

Energetic quantities

Energy loss in the forced response

SDOF
energetics

■ Then:

$$\begin{aligned}\mathcal{E}_d^f(+\infty) &= \int_{\mathbb{R}} D(\dot{u}^f(t))^2 dt \\ &= \frac{D}{2\pi} \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 |\hat{f}(\omega)|^2 d\omega \quad (\text{Plancherel}) \\ &\simeq \frac{D}{2\pi} |\hat{f}(\omega_p)|^2 \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 d\omega \quad (\text{using (i)-(ii)}) \\ &= \frac{|\hat{f}(\omega_p)|^2}{2M},\end{aligned}$$

and the overall dissipated energy is independent of the damping.¹

■ It is related to the overall mechanical energy by:

$$\mathcal{E}_d^f(+\infty) \simeq \int_{\mathbb{R}} \eta_p \omega_p \mathcal{E}^f(t) dt.$$

¹In 3D turbulent flows at high Reynolds number, the dissipation rate is independent of the viscosity.

SDOF
oscillator
(reminder)
Notations
Energetic
quantities
Stationary
excitation
Stationary
forced
response
Equipartition
Stationary
evolutionary
response
Summary

Energetic quantities

Evolutionary response

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response

Equipartition

Stationary
evolutionary
response

Summary

- The power balance integrated between $t = 0$ and $t = +\infty$:

$$\mathcal{E}_d(+\infty) = \mathcal{E}_0 + \mathcal{E}_{\text{IN}}(+\infty).$$

Outline

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)
Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition
Stationary
evolutionary
response

Summary

1 SDOF oscillator (reminder)

- Notations
- Energetic quantities

2 Stationary excitation

- Stationary forced response
- Equipartition
- Stationary evolutionary response

Stationary excitation

Forced response of the SDOF oscillator

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)
Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition
Stationary
evolutionary
response

Summary

- **Data:** $(F_t, t \in \mathbb{R})$ is a $\mathbb{R}$ -valued second order, centered stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, and mean-square stationary.
- **Hypothesis:** $\exists \omega \mapsto S_F(\omega) : \mathbb{R} \rightarrow \mathbb{R}_+$ even, integrable.

Proposition

- $(U_t^f, t \in \mathbb{R})$ is a $\mathbb{R}$ -valued second order, centered stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}$, and mean-square stationary s.t. $S_U(\omega) = |\hat{h}(\omega)|^2 S_F(\omega)$.
- The same holds for its mean-square derivatives $(\dot{U}_t^f, t \in \mathbb{R})$ and $(\ddot{U}_t^f, t \in \mathbb{R})$, with $S_{\dot{U}}(\omega) = \omega^2 S_U(\omega)$ and $S_{\ddot{U}}(\omega) = \omega^4 S_U(\omega)$, respectively.

Stationary excitation

Energetics of the stationary forced response

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition
Stationary
evolutionary
response

Summary

- The instantaneous power balance reads:

$$\dot{\mathcal{E}}_t^f = \Pi_{\text{IN},t} - \Pi_{\text{d},t}^f,$$

as an equality of second-order random variables.

- Considering the mathematical expectation with:

$$\mathbb{E}\{(U_t^f)^2\} = R_U(t=0) = \text{Constant independent of } t,$$

$$\mathbb{E}\{(\dot{U}_t^f)^2\} = -\frac{d^2 R_U}{dt^2}(t=0) = \text{Constant independent of } t,$$

yields $\mathbb{E}\{\mathcal{E}_t^f\} = \text{Constant}$, and $\mathbb{E}\{\dot{\mathcal{E}}_t^f\} = 0$. Hence:

$$\mathbb{E}\{\Pi_{\text{IN},t}\} = \mathbb{E}\{\Pi_{\text{d},t}^f\}.$$

Stationary excitation

Equipartition in the stationary forced response

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response

Equipartition

Stationary
evolutionary
response

Summary

■ Hypotheses (wideband excitation):

■ i $\xi_p \ll 1$;

■ ii $\omega \mapsto S_F(\omega)$ varies slowly on $[-\omega_p - \frac{b_e}{2}, \omega_p + \frac{b_e}{2}]$.

■ Then:

$$\begin{aligned}\mathbb{E}\{\mathcal{E}_{c,t}^f\} &= \frac{M}{2} \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 S_F(\omega) d\omega \\ &\simeq \frac{M}{2} S_F(\omega_p) \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 d\omega \quad (\text{using (i)-(ii)}) \\ &= \frac{\pi S_F(\omega_p)}{2D}\end{aligned}$$

- Likewise $\mathbb{E}\{\mathcal{E}_{p,t}^f\} \simeq \frac{\pi S_F(\omega_p)}{2D}$, independently of the mass or the stiffness, s.t.

$$\boxed{\mathbb{E}\{\mathcal{E}_t^f\} \simeq \frac{\pi S_F(\omega_p)}{D}}.$$

Stationary excitation

Equipartition in the stationary forced response (extended proof)

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition

Stationary
evolutionary
response

Summary

$$\begin{aligned}\mathbb{E}\{\mathcal{E}_{c,t}^f\} &= \frac{M}{2} \mathbb{E}\{(\dot{U}_t^f)^2\} \\&= \frac{M}{2} R_{\dot{U}}(0) \quad (\mathbb{E}\{\dot{U}_t \dot{U}_{t'}\} = R_{\dot{U}}(t - t') \implies \mathbb{E}\{(\dot{U}_t)^2\} = R_{\dot{U}}(t - t) = R_{\dot{U}}(0)) \\&= \frac{M}{2} \int_{\mathbb{R}} S_{\dot{U}}(\omega) d\omega \quad (R_{\dot{U}}(t) = \int_{\mathbb{R}} e^{i\omega t} S_{\dot{U}}(\omega) d\omega \implies R_{\dot{U}}(0) = \int_{\mathbb{R}} S_{\dot{U}}(\omega) d\omega) \\&= \frac{M}{2} \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 S_F(\omega) d\omega \quad (S_{\dot{U}}(\omega) = \omega^2 S_U(\omega) = \omega^2 |\hat{h}(\omega)|^2 S_F(\omega)) \\&\simeq \frac{M}{2} S_F(\omega_p) \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 d\omega \quad (\text{using (i)-(ii)}) \\&= \frac{\pi S_F(\omega_p)}{2D} \quad (\int_0^{+\infty} \omega^2 |\hat{h}(\omega)|^2 d\omega = \frac{\pi}{2DM} = \frac{1}{2} \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 d\omega)\end{aligned}$$

Stationary excitation

Energy loss in the stationary forced response

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition

Stationary
evolutionary
response

Summary

- Then:

$$\begin{aligned}\mathbb{E}\{\Pi_{d,t}^f\} &= D \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 S_F(\omega) d\omega \\ &\simeq D S_F(\omega_p) \int_{\mathbb{R}} \omega^2 |\hat{h}(\omega)|^2 d\omega \quad (\text{using (i)-(ii)}) \\ &= \frac{\pi S_F(\omega_p)}{M},\end{aligned}$$

and the average dissipated power is independent of the damping.

- It is related to the average mechanical energy by:

$$\boxed{\mathbb{E}\{\Pi_{d,t}^f\} \simeq \eta_p \omega_p \mathbb{E}\{\mathcal{E}_t^f\}}.$$

Stationary excitation

Stationary evolutionary response

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition

Stationary
evolutionary
response

Summary

- Assuming null initial conditions to simplify, the evolutionary response $(U_t, t \geq 0)$ is a $\mathbb{R}$ -valued second-order, centered stochastic process defined on $(\Omega, \mathcal{F}, P)$, indexed on $\mathbb{R}_+$, which is non stationary:

$$U_t = \int_0^t \mathbb{h}(t - \tau) F_\tau \, d\tau .$$

- **Property:** if $D > 0$, $\lim_{t \rightarrow +\infty} \|U_t - U_t^f\| = 0$.

Summary

SDOF
energetics

É. Savin

SDOF
oscillator
(reminder)

Notations
Energetic
quantities

Stationary
excitation

Stationary
forced
response
Equipartition

Stationary
evolutionary
response

Summary

- "Equivalence" between:
 - 1 The overall energetic quantities for the forced response to a deterministic, wideband excitation;
 - 2 The average energetic quantities for the stationary forced response to a random, wideband (m.s.) stationary excitation;
 - 3 The time average energetic quantities for the forced response of the randomized SDOF oscillator to an harmonic excitation.
- These different cases are often (unduly) merged in the structural-acoustics literature.
- **Outlook:** multiple DOF systems.